

Chapter 4, Section 5

James Lee

April 24, 2026

1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\text{Aut } X$ of automorphisms of X is finite.

Proof. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g_3^1 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbf{P}^1$, ramified at $0, 1, \infty$, and $2g - 1 = 7$ additional points, up to the action of a certain finite group. If X is a nonhyperelliptic curve of genus 4, then its canonical embedding in $\mathbf{P}^3$ is the complete intersection of a unique irreducible quadric surface and an irreducible cubic surface. Conversely, if X is a nonsingular curve in $\mathbf{P}^3$ which is a complete intersection of a quadric and a cubic surface, then X is a canonical curve of genus 4. The family of all these curves is parametrized by an open subset $U \subseteq |\mathcal{O}_{\mathbf{P}^3}(2)| \times |\mathcal{O}_{\mathbf{P}^3}(3)|$, with $\dim U = 9 + 19 = 28$, because a form of degree 2 has 10 coefficients, and a form of degree 3 has 20 coefficients. So there is a morphism $U \rightarrow \mathcal{M}_4$ given by $(Q, F_3) \mapsto Q \cap F_3$, and it remains to count the dimension of each fibre, so let U_X be any fibre, with $X \in \mathcal{M}_g$. First consider the action of $\text{PGL}(3)$ on $\mathbf{P}^3$. It has dimension $(3 + 1)^2 - 1 = 15$, and given any pair $(Q, F_3) \in U_X$, we have $Q \cap F_3 \cong \sigma(Q \cap F_3) \cong \sigma(Q) \cap \sigma(F_3)$ for all $\sigma \in \text{PGL}(3)$. Since the embedding is canonical, two of them are isomorphic as abstract curves if and only if they differ by an automorphism of $\mathbf{P}^3$.

More specifically, if $C = Q \cap F_3$ and $C' = Q' \cap F'_3$ are isomorphic, then there exists $\sigma \in \text{PGL}(3)$ such that $C' = \sigma(C)$, which implies $Q' \cap F'_3 = \sigma(Q) \cap \sigma(F_3)$, so if two genus 4 curves are isomorphic, then they lie in the same quadric. Since a genus 4 curve in $\mathbf{P}^3$ is contained in a unique quadric, we have $Q' = \sigma(Q)$. For each quadric Q , we consider the number of unique genus 3 curves $Q \cap F_3$.

A nonhyperelliptic genus 4 curve has only one g_3^1 if and only if it lies on a singular quadric surface in its canonical embedding, and the space of singular quadrics form an irreducible family of dimension 8 in $|\mathcal{O}_{\mathbf{P}^3}(2)|$. □

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in $\mathbf{P}^4$ is a complete intersection $F_1.F_2.F_3$ form a family of dimension 12.
- (b) X has a g_3^1 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbf{P}^2 \rightarrow \mathbf{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbf{P}^4$, with V isomorphic to $\mathbf{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g_3^1 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X . Thus V and g_3^1 are unique.

Proof. □

- (a)
- (b)
- (c)

7. (a) Any automorphism of a curve of genus 3 is induced by an automorphism of $\mathbf{P}^2$ via the canonical embedding.
- (b) Assume $\text{char } k \neq 3$. If X is the curve given by

$$x^3y + y^3z + z^3x = 0,$$

the group $\text{Aut } X$ is the simple group of order 168, whose order is the maximum $84(g - 1)$ allowed by (Ex. 2.5).

(c) Most curves of genus 3 have no automorphisms except the identity.

Proof.

(a)

(b)

(c)

□